

StreamPay

Blockchain Fintech Platform

PROJECT TYPE Blockchain Fintech Platform	PLATFORM Responsive web application and worker PWA
AUDIENCE Employers, workers, and payroll teams	PRIMARY FOCUS Per-second salary streaming on Stellar

Overview

An on-chain salary streaming platform that lets employers fully escrow a pay period in a Soroban smart contract while worker wages accrue automatically from ledger time. Workers can cash out earned amounts at any time, and fair cancellation pays every earned stroop before returning the unearned balance. StreamPay combines a responsive employer dashboard, a mobile-first worker PWA, and public Stellar testnet receipts.

Core Features

- Full-pay-period escrow through a Soroban smart contract on Stellar testnet
- Ledger-timestamp accrual with integer-only stroop calculations
- Any-time worker cash-outs with no minimum withdrawal amount
- Fair cancellation: earned wages to the worker, remainder to the employer
- Freightier-signed employer workflow for creating and monitoring streams
- Mobile-first worker PWA with a live ticking balance and currency views
- Encrypted in-app testnet wallet, PIN lock, backup, and privacy controls
- Public explorer receipts, CSV exports, and lifecycle-aware dashboards

Experience and Design

- A single landing page separates employer and worker paths without requiring blockchain knowledge.
- Employer dashboards summarize active streams, money in flight, deposits, filters, and CSV exports.
- The phone-first worker interface shows available pay, live per-second earnings, receipts, and activity.
- Plain-language testnet notices, local currency views, privacy controls, and lifecycle badges keep status clear.

Architecture and Data

- React 19, TypeScript, Vite, and Tailwind CSS 4 deliver one responsive web application.
- A Rust Soroban contract is the backend; no application server or database mediates balances.
- Stellar SDK reads contract state, submits invocations, and links transactions to public explorer proof.
- Ledger timestamps drive accrual while the UI smooths the ticker between authoritative chain reads.

Security and Privacy

- Freightier signs employer transactions without exposing the wallet secret to the application.
- Worker signing uses an encrypted local testnet keypair protected by a six-digit device PIN and backup flow.
- Contract authorization restricts create and cancel to employers and withdrawals to the designated worker.
- No admin, pause, drain, or upgrade surface exists; full escrow and exact conservation are contract invariants.

Technology Stack

Rust	Smart contract language
Soroban SDK	Stellar smart contract framework
Stellar SDK	Contract reads and transaction submission
React 19	Application framework and UI
TypeScript	Application language and typed chain integration
Vite	Build tooling and local development
Tailwind CSS 4	Interface styling and design system
Freighter API	Employer wallet connection and signing

Performance and Reliability

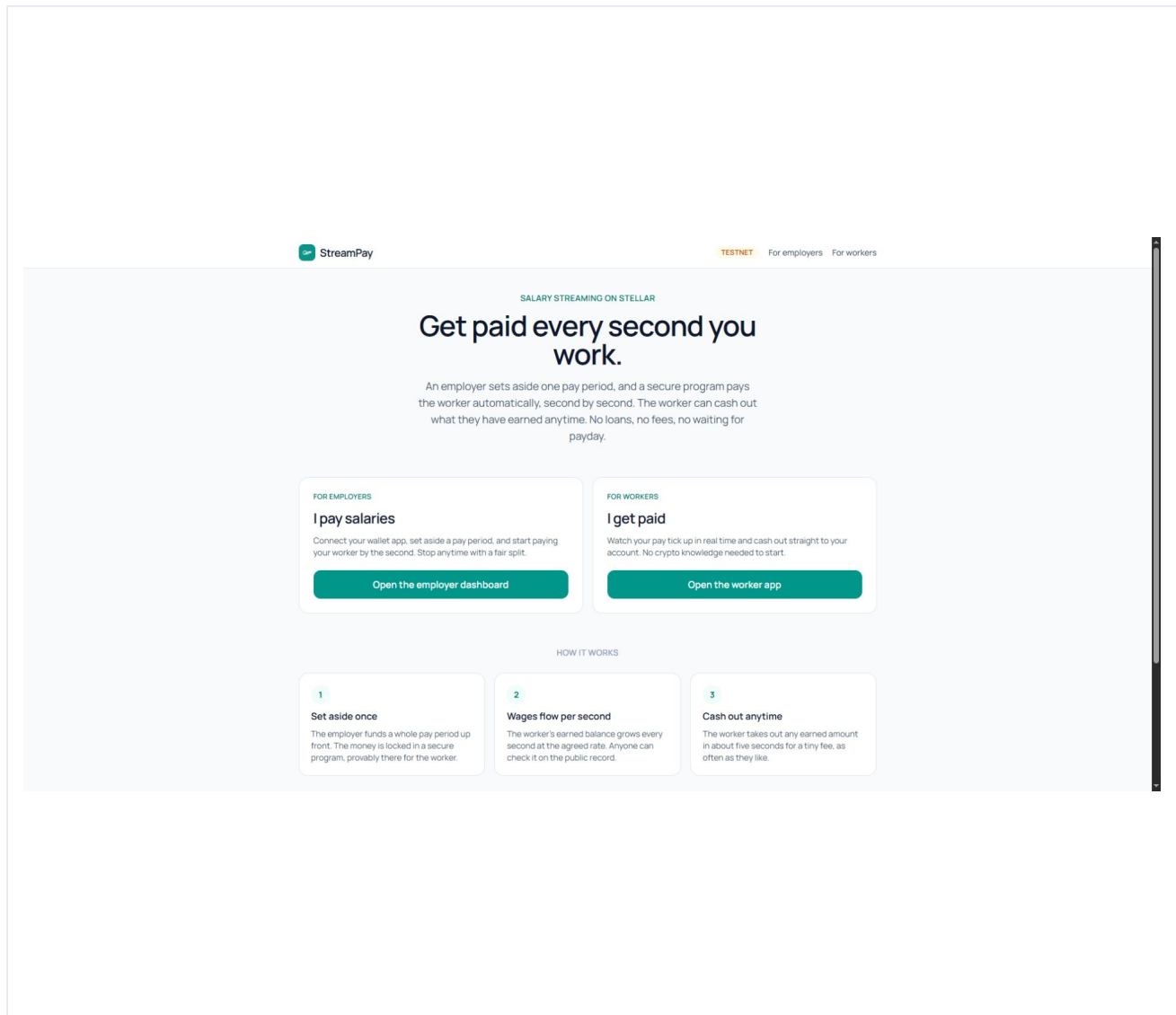
- Six-second chain polling refreshes authoritative balances while the main ticker animates each second.
- Integer-only stroop math prevents floating-point drift in accrual, withdrawals, and cancellation splits.
- Fully escrowed streams cannot run dry mid-period, and earned balances remain withdrawable after vesting.
- Automated contract properties and web tests cover conservation, authorization, lifecycle, and edge cases.

Platform Requirements

- Current Chrome, Edge, Firefox, or Safari browser with JavaScript enabled.
- Stable internet connection to Stellar testnet RPC and public explorer services.
- Freighter on Stellar testnet for employer-created and employer-cancelled transactions.
- Worker experience supports mobile browsers and installable PWA presentation; all balances are test tokens.

PAGE OVERVIEW

Landing Page



Landing Page

Interface view from StreamPay.

PAGE OVERVIEW

Employer Dashboard

The screenshot displays the StreamPay Employer Dashboard. At the top, the StreamPay logo and 'TESTNET' label are on the left, while navigation links for 'Dashboard', 'Create stream', 'Settings', a user profile 'GAKE . . . LUFA', and 'Sign out' are on the right. The main section is titled 'Dashboard' with a subtext 'Set aside a pay period once; wages flow to your workers every second.' and a 'Create stream' button. Below this are three summary cards: 'ACTIVE STREAMS' showing 1 active stream, 'STREAMING NOW' showing 4,000 XLM being streamed, and 'TOTAL DEPOSITED' showing 4,000 XLM deposited. A filter bar shows 'All' selected, with options for 'Active', 'Completed', and 'Cancelled'. A search bar and 'Export CSV' button are also present. The 'Workers you are paying' section features a card for 'Stream #13' (Active), showing worker 'GAUN . . . 3RTE', a duration of 2h 30m, and a progress bar indicating 2% paid out. It also shows 'Set aside 4,000 XLM' and 'Earned so far 87.11 XLM'. At the bottom of the card are links for 'Copy worker link', 'Public receipt', and a 'Stop paying' button.

StreamPay TESTNET Dashboard Create stream Settings GAKE . . . LUFA Sign out

Dashboard

Set aside a pay period once; wages flow to your workers every second.

Create stream

ACTIVE STREAMS

1

Currently paying a worker

STREAMING NOW

4,000 XLM

Money in flight across active streams

TOTAL DEPOSITED

4,000 XLM

Across every stream you have funded

All Active Completed Cancelled

Search name or address Export CSV

Workers you are paying

Stream #13 Active

Worker	Set aside
GAUN . . . 3RTE	4,000 XLM
Duration	Earned so far
2h 30m	87.11 XLM

2% paid out

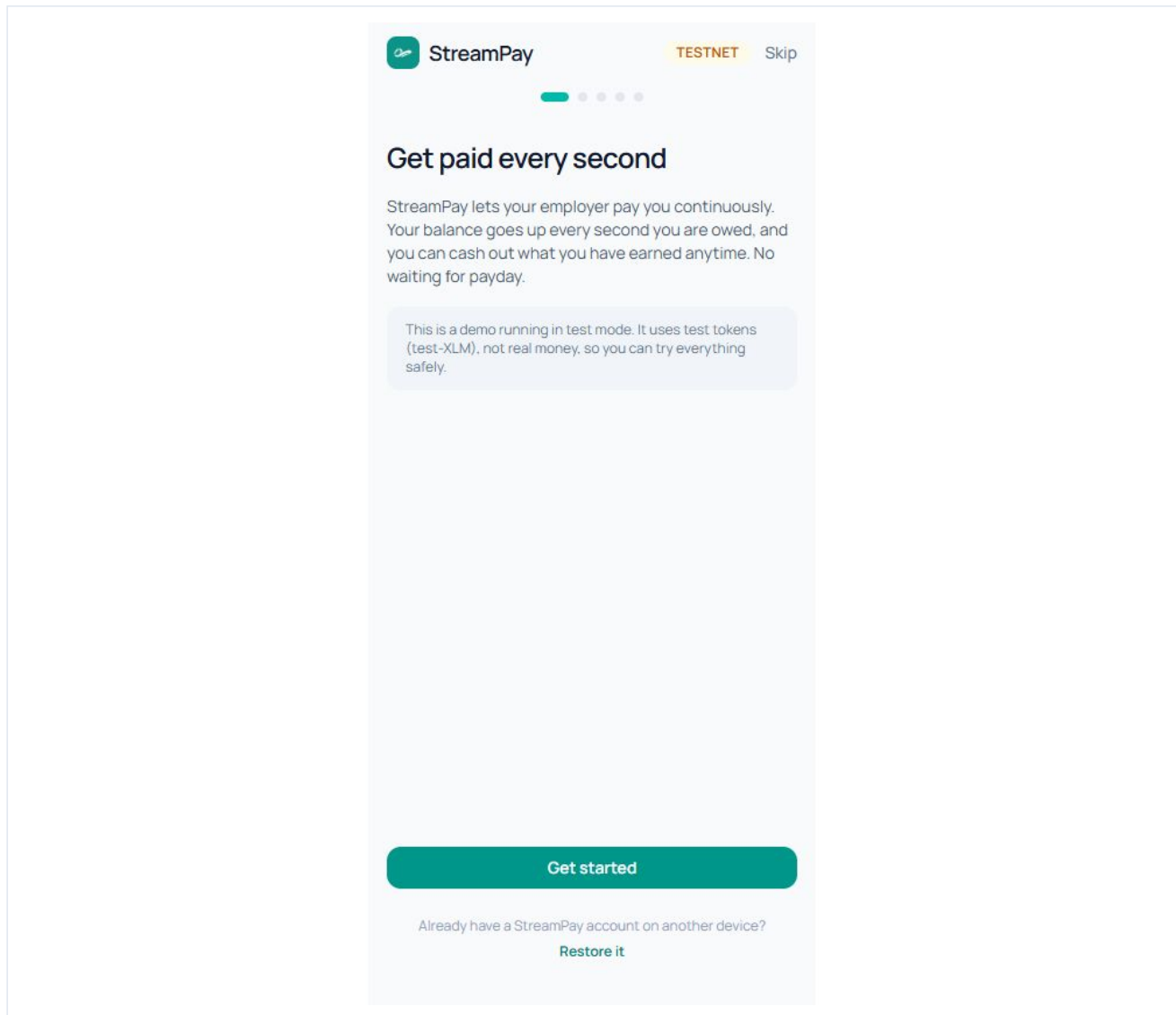
Copy worker link Public receipt Stop paying

Employer Dashboard

Interface view from StreamPay.

PAGE OVERVIEW

Worker Onboarding

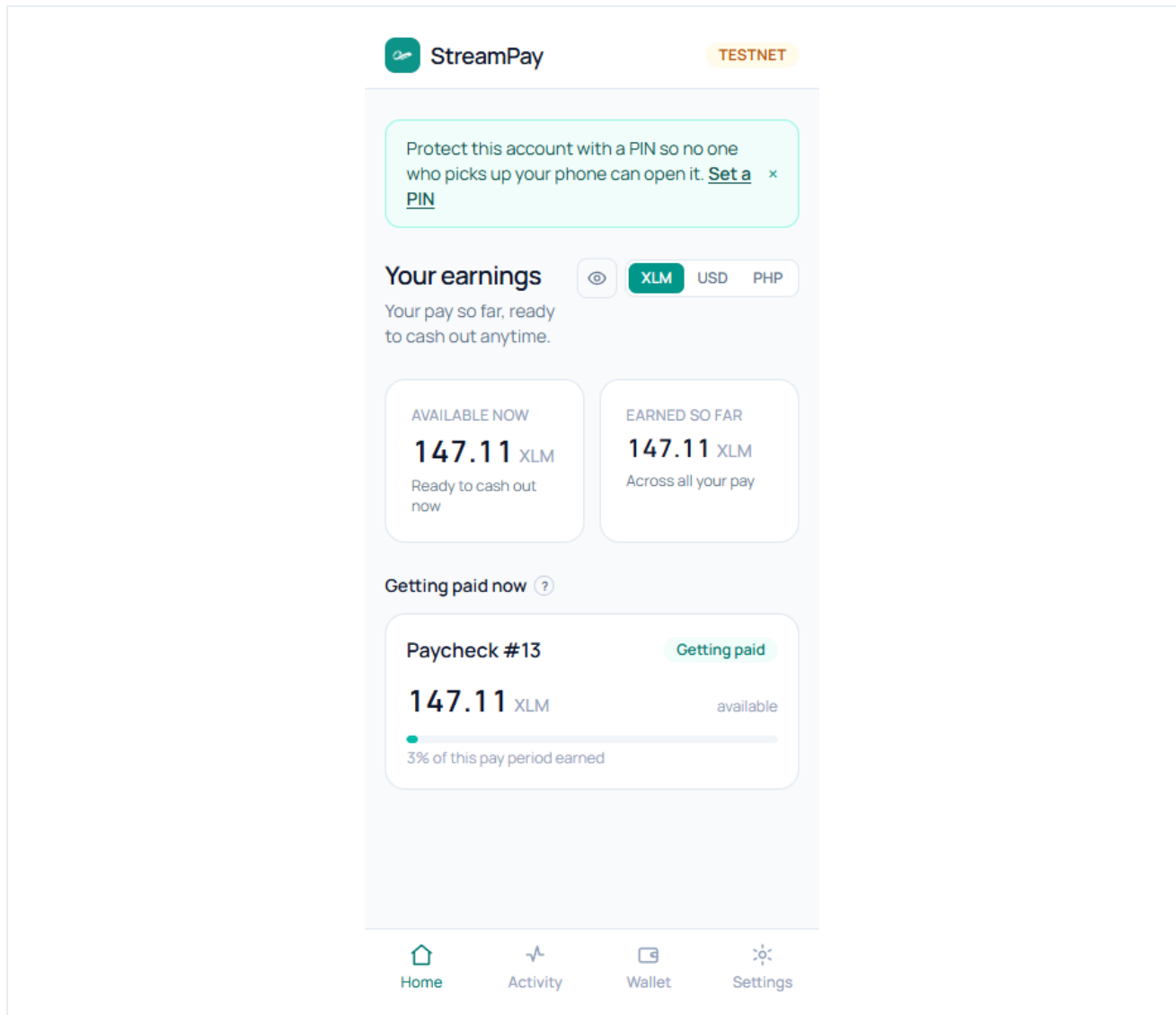


Worker Onboarding

Interface view from StreamPay.

PAGE OVERVIEW

Worker Earnings Home

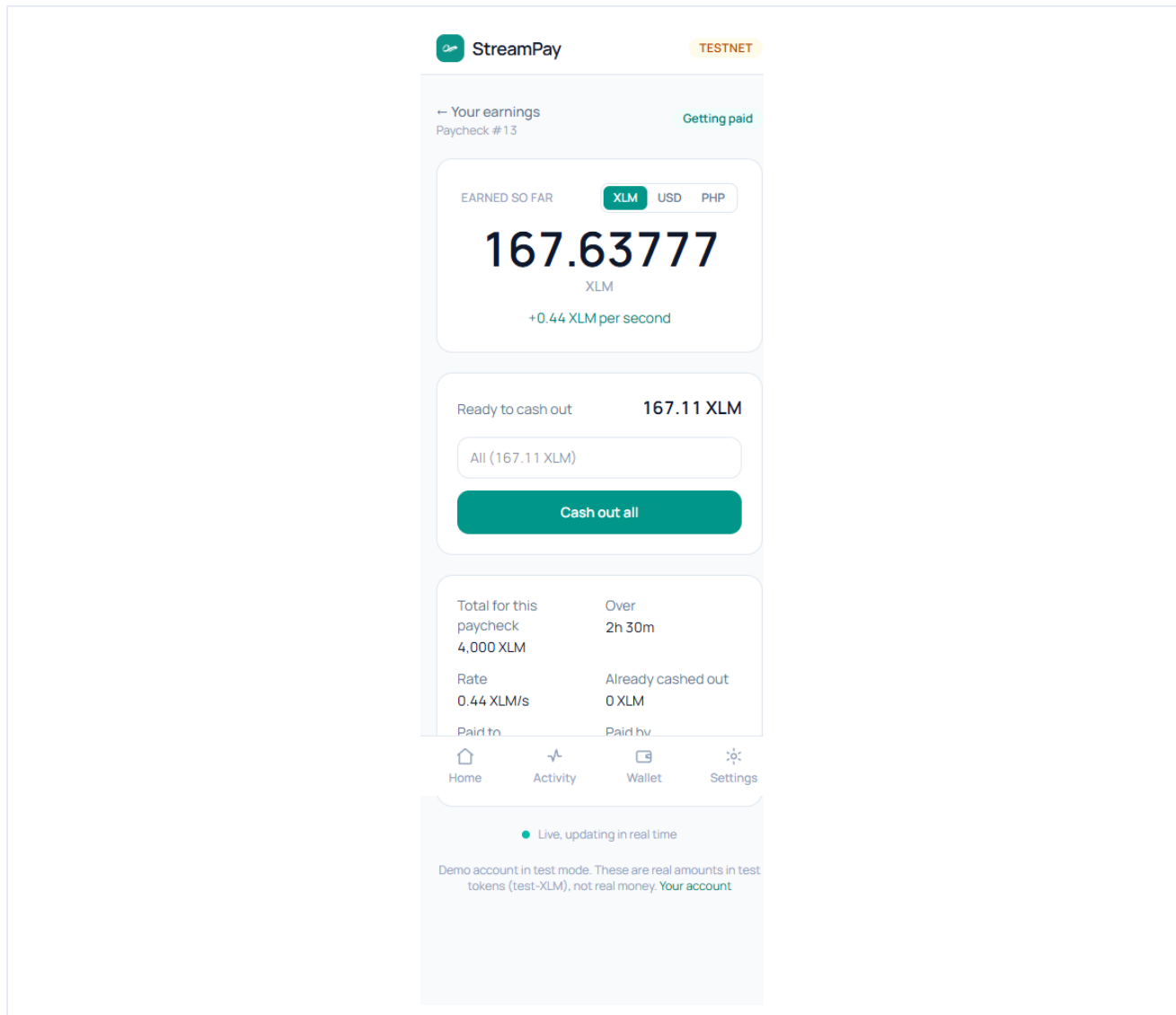


Worker Earnings Home

Interface view from StreamPay.

PAGE OVERVIEW

Live Paycheck and Cash Out

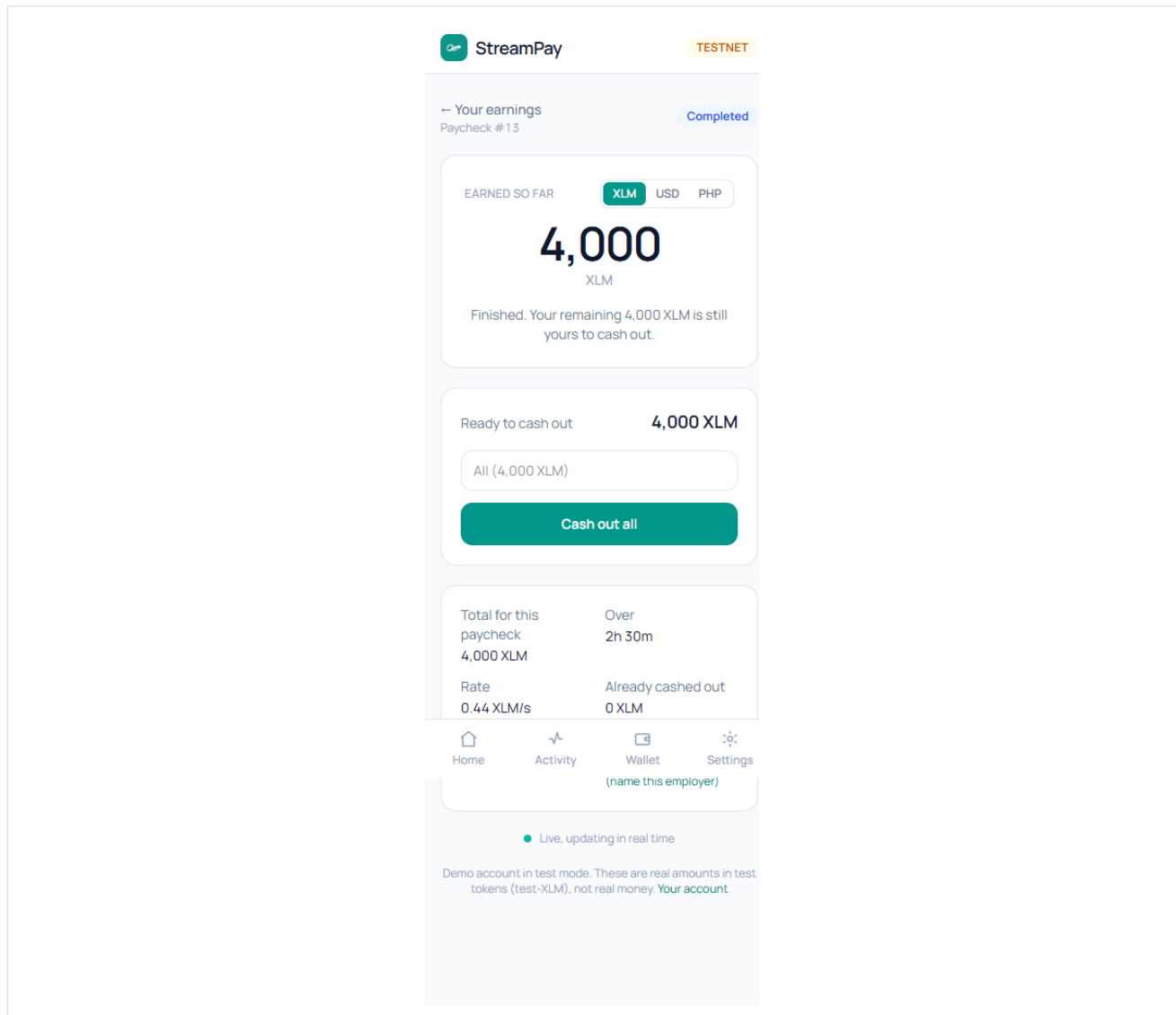


Live Paycheck and Cash Out

Interface view from StreamPay.

PAGE OVERVIEW

Completed Paycheck

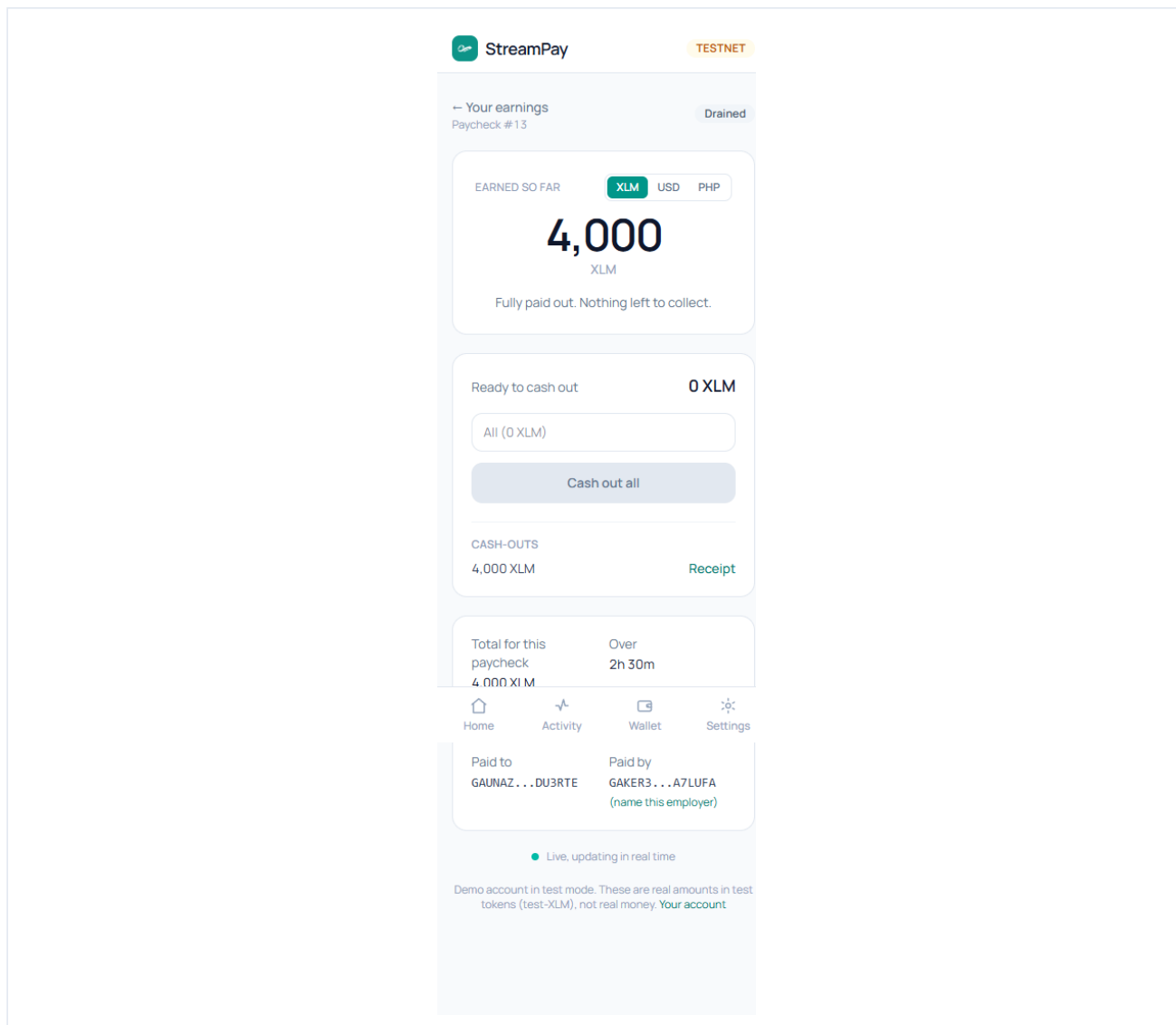


Completed Paycheck

Interface view from StreamPay.

PAGE OVERVIEW

Completed Cash Out

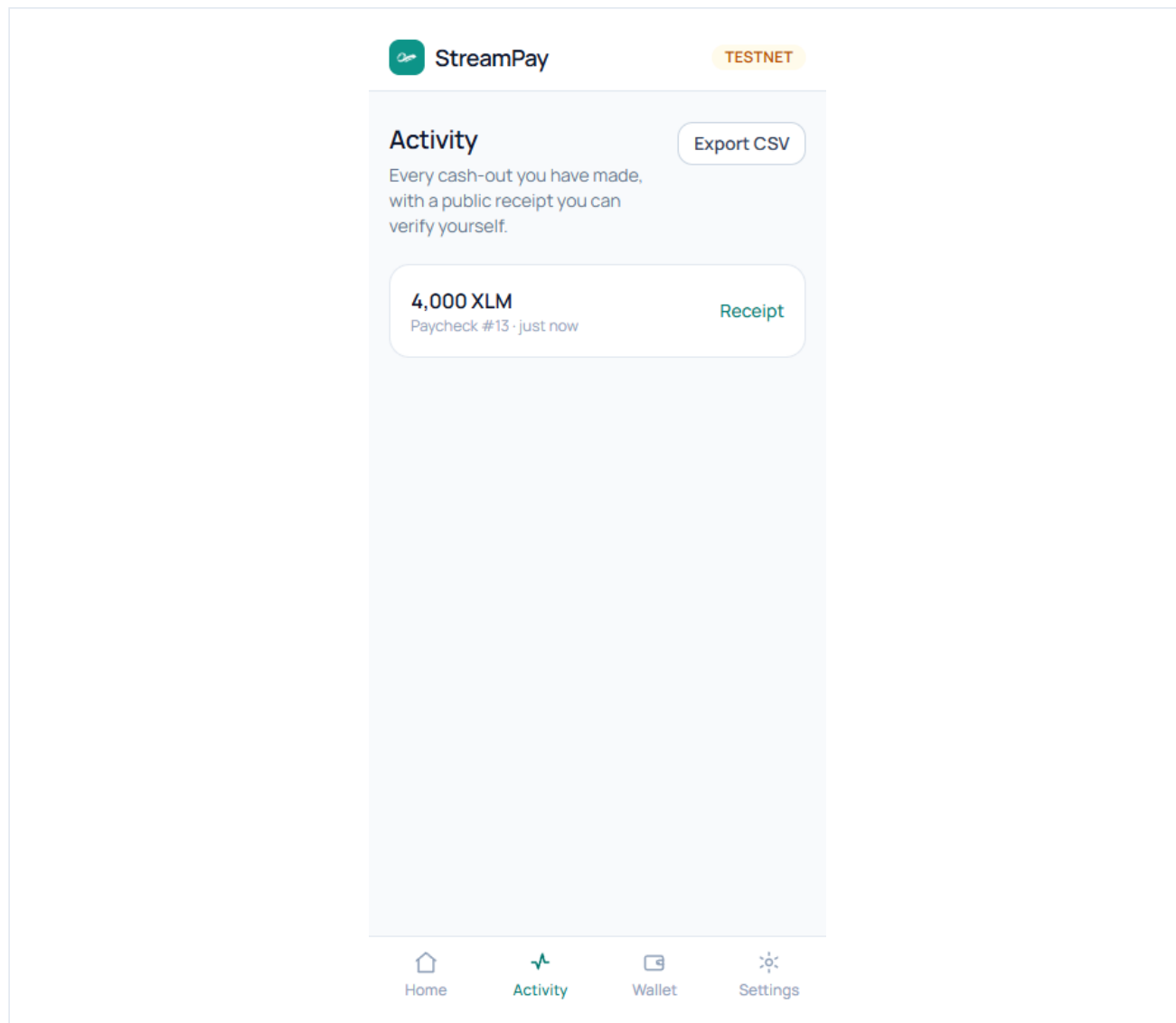


Completed Cash Out

Interface view from StreamPay.

PAGE OVERVIEW

Cash-Out Activity



Cash-Out Activity

Interface view from StreamPay.